Higher National Diploma in Engineering (Electrical)
4th Year, First Semester Examination – 2021
EE 3214 – Construction and Protection of Electrical systems

Instructions for Candidates:	No. of questions:06
Answer Any **Five (05)** questions.	No. of pages : 03
Each question carries equal marks	Time : **03 Hours**

Q1.

a) Write five advantages of Per Unit System in an electrical power system? (05 Marks)

b) A 10 MVA, 6·6 kV, 3-phase star-connected alternator having a reactance of 20% is connected through a 5 MVA, 6·6 kV/33 kV transformer of 10% reactance to a transmission line having a resistance and reactance per conductor per kilometer of 0·2 Ω and 1 Ω respectively. 50 km along the line, a short-circuit occurs between the three conductors (symmetrical fault). Find the current fed to the fault by the generator. (15Marks)

[Total 20 Marks]

Q2.

a) Name five types of Common faults in three-phase system (05 Marks)

b) The symmetrical components of red phase voltage in a 3-phase unbalanced system are $V_{R0} = 10\angle 180$ V, $V_{R1} = 50\angle 0$ V and $V_{R2} = 20\angle 90$ V. Determine the phase voltages of V_R, V_Y, and V_B. (15 Marks)

[Total 20 Marks]

Q3.

a) Explain the operation principle of Relay, trip circuit, and circuit breaker in electrical protection system. (5 Marks)

b) Explain the construction and operation of induction-type directional and non-direction over current relays with neat sketch diagrams? (08+07 Marks)

[Total 20 Marks]

Q4.

a) Explain the operation of the differential current protection system for the generator with a neat diagram. (05 Marks)

a) A star connected 11 kV, 100 MVA alternator is grounded through a resistance of 10 ohms. The current transformers have a ratio of 1000/5. The relay is set to operate when there is an out-of-balance current of 0.5 A. Find the percentage of the generator winding protected by the differential protection system when the machine is operating at its normal voltage.

(15 Marks)

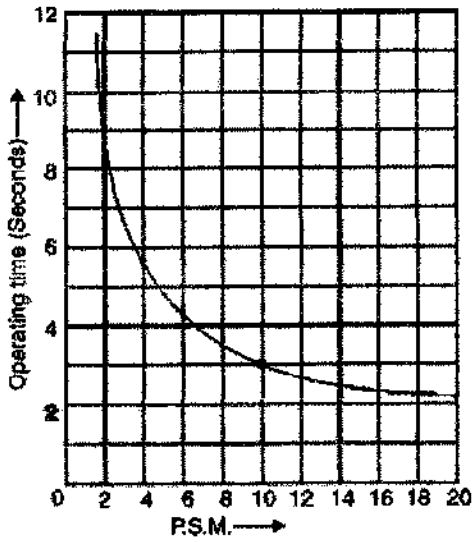
[Total 20 Marks]

Q5.

a) Write short notes on the following related to the protective relay?

I. Pick up current
II. Current Setting
III. Plug Setting Multiplier (P.S.M)
IV. Time Setting Multiplier (T.S.M)
V. Time / P.S.M curve (05 marks)

b) Determine the time of operation of a 5-ampere, 4-second overcurrent relay having a current setting of 150% and a time setting multiplier of 0·6 connected to the supply circuit through a 400/5 current transformer when the circuit carries a fault current of 4500 A. (15 Marks)



Q5.b

[Total 20 Marks]

Q6.

a) Describe the construction & working principle of the Buchholz relay? (05 Marks)

b) A 3ϕ, transformer having a line voltage ratio of 300 V/11 kV is protected by Merz-Price circulating current system. The transformer is connected to star delta and the protective transformer on the 300 V side has a current transformer ratio of 200/5.

i. Draw the arrangement of the Merz-Price circulating current protection system for the transformer. (5 Marks)

ii. What must be the ratio of the protective transformers on the 11kV side? (10 Marks)

[Total 20 Marks]